

Chapter 9

Radiation Therapy in Ocular Melanoma



Andrew J. Wong and Bin S. Teh

Significance of Radiotherapy in Uveal Melanoma

Radiotherapy plays an important role in uveal melanoma (UM) treatment. Historically local treatment of UM was managed with enucleation. A series of Collaborative Ocular Melanoma Study (COMS) trials established eye plaque brachytherapy as a standard treatment option for most UMs, demonstrating no significant difference in outcome metrics while preserving the globe and visual function. Numerous external beam radiotherapy modalities and techniques (photon-fractionated and stereotactic radiosurgery, proton and particle-based radiotherapy) further complement plaque brachytherapy with an array of definitive or adjunct treatment options.

A fundamental balancing act for radiotherapy in UM is achieving both excellent tumor control and sparing nearby critical structures to preserve vision and maintain quality of life. Local recurrence in UM increases the risk of metastasis and disease-specific mortality [1, 2]. Thus, an understanding of the different radiotherapy modalities and factors to optimize local control has impactful ramifications.

A. J. Wong

Department of Radiation Oncology, University of Texas Medical Branch Galveston,
Galveston, TX, USA

e-mail: ajwong@utmb.edu

B. S. Teh (✉)

Department of Radiation Oncology, Houston Methodist Hospital, Houston, TX, USA

e-mail: bteh@houstonmethodist.org

Indications for Radiotherapy

In this chapter, the COMS staging is the basis for guiding the radiation treatment paradigm approach based on tumor size. This staging classification was utilized in a series of key trials that established treatment approaches and baseline outcome characteristics. In particular, a COMS small tumor is defined as between 1 and 3 mm in height and between 5 and 16 mm in basal diameter, a medium tumor is 3.1–8 mm in height and no more than 16 mm in basal diameter, and a large tumor is more than 8 mm in height or more than 16 mm across. The American Joint Committee on Cancer (AJCC) 8th edition staging (2017) utilizes tumor dimensions in addition to presence of ciliary body involvement and extraocular extension to describe a tumor size category. Prognostic stage groups are determined by tumor size category in addition to presence of nodal or distant metastasis [3]. Please refer to the AJCC staging manual for details.

Radiotherapy for Small Versus Medium Versus Large Tumors

Small, melanocytic tumors with an uncertain diagnosis for uveal melanoma or with few risk factors for growth can be observed [4]. Two-thirds of such lesions may represent choroidal nevi and do not grow, with approximately one-third of lesions progressing at 5 years [5]. Observation is particularly applicable for small tumors near the fovea, optic nerve, or critical structures in which radiation-induced vision toxicity is likely. Radiation therapy is indicated for small tumors when significant growth is demonstrated after conservative management – observation or local treatments such as local resection, photodynamic laser photocoagulation, or transpupillary thermotherapy.

For medium-sized tumors, eye plaque brachytherapy, particle/proton radiotherapy, and enucleation are all options. COMS report No. 28 was a prospective randomized trial that compared enucleation and iodine-125 eye plaque brachytherapy for select cohort of medium-sized choroidal melanomas. It found no difference in survival rates through 12 years of follow-up [6]. These results established eye plaque brachytherapy as the standard treatment approach for most medium-sized tumors, allowing for equivalent control while sparing both globe and vision.

Enucleation and particle/proton radiotherapy are options for local treatment to large tumors. If extraocular extension is present at the time of enucleation, particle beam or photon beam to the orbit can be considered. Of note, neoadjuvant external beam radiation therapy preceding enucleation did not improve outcomes over enucleation alone [7].

Local Recurrence and Metastatic Disease

Plaque brachytherapy and particle beam therapy are definitive options for treating intraocular recurrences. Re-irradiation can be considered for recurrent lesions in the

absence of metastatic disease. In the setting of limited or symptomatic hepatic disease, photon-based radiotherapy can be utilized for palliation.

What Are Other Options Preferred Over Radiotherapy?

Conservative management (observation, local treatment) is preferred for small COMS tumors as described above. Enucleation is considered for the following: salvage setting, large tumors involving substantial intraocular volume, extrascleral extension, nonfunctional eye, compromised (e.g., marked neovascularization) or symptomatic (e.g., painful) eye, or poor predicted functional outcome from other treatment options.

Radiotherapy Modalities

Episcleral Plaque Brachytherapy

Indication/Contraindications

Episcleral plaque brachytherapy (EPB) is a common, effective, vision-sparing method for treating uveal melanoma. Applications for plaque therapy have expanded to include most uveal melanoma of varying sites (including iris, ciliary body, choroid, subfovea, juxtapapillary, circumpapillary) and sizes (small and large tumors as well as those with limited extrascleral extension (i.e., AJCC T1, T2, T3, T4a-d)) [8]. EPB is appropriate for tumors ≤ 18 mm in largest base diameter or ≤ 10 mm in thickness as definitive therapy [9] and for small tumors that progress after observation or local recurrence after a prior local therapy. EPB is contraindicated for large tumors with extraocular extension (AJCC T4e) or with dimensions that exceed the size limit of brachytherapy, in blind painful eyes, and in cases with no light perception vision [8].

Overview of Treatment Planning and Delivery

Performing eye plaque brachytherapy is a multidisciplinary team effort involving an ocular oncologist, radiation oncologist, and brachytherapy physicist.

Preoperative plaque treatment planning takes into consideration several patient, tumor, and radiotherapy characteristics. A fundus diagram describing geometry and orientation of tumor within the eye, derived from ophthalmic examination, ultrasound findings, and photographic images (CT or MRI), is utilized as the basis for treatment planning. The appropriate radionuclide, prescription dose, dose rate, and eye plaque are selected for optimal treatment delivery as determined by a computerized treatment planning system. See Fig. 9.1 for representative images in plaque

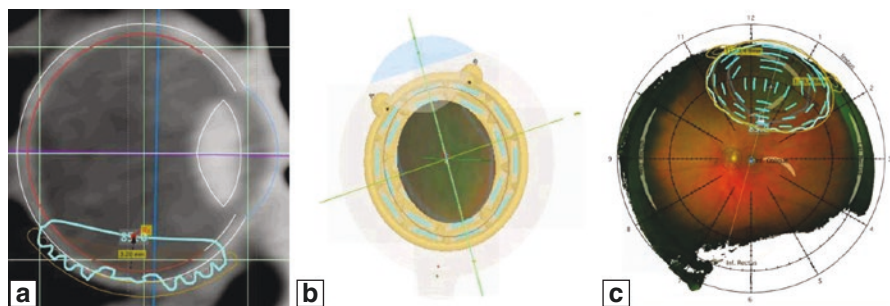


Fig. 9.1 3D treatment planning of plaque brachytherapy using plaque simulator software. (a) Fusion of thin-cut CT image set to 2D and 3D models of the affected eye, plaque, tumor, and dose distribution. (b) 3D reconstruction of the eye with plaque placed in relation to critical structures, including optic nerve and lens. (c) Retinal diagram with plaque, its coordinates, and isodose distribution. *CT* computed tomography, *2D* two-dimensional, *3D* three-dimensional

treatment planning. Eye plaques may be gold-plated with grooves to seat radiation sources (iodine-125, palladium-103, cesium-131) or solid beta-emitting plaques (ruthenium-106, strontium-90) [10]. Eye plaques can be customized by shape and size to accommodate sparing of important visual structures. For instance, notched plaques can be utilized for peripapillary tumors.

Particulars of plaque surgery are detailed in the corresponding chapter in this text. Under regional or general anesthesia, tumor localization is performed using transillumination, indirect ophthalmoscopy, or light pipe diathermy. A safety margin is often incorporated about the target volume. A dummy plaque may be used to verify position, followed by plaque placement and suturing. Imaged guidance may be utilized at the time of plaque placement to validate full spatial coverage of the target. The tumor is treated for a predetermined set period (typically 4–7 days) for which a lead eye shield is worn, followed by plaque removal under regional or general anesthesia.

Dose Prescription Considerations

For COMS iodine-125 plaques, the standard dose prescription is established as 85 Gy to the tumor apex (or 5 mm from internal surface of sclera if height is <5 mm) with ≥ 2 mm margin around tumor edge if possible, at a dose rate of 0.6–1.05 Gy/hr. The largest commercially available eye plaque is 22 mm in diameter. Factoring in a 2 mm margin on the tumor edge, generally the maximum allowed tumor dimensions for an eye plaque are 18 mm basal diameter and 10 mm thickness (8 mm if optic nerve is involved). For other radioisotopes, including non-COMS iodine-125 plaques, there are variations of dose (generally 60–100 Gy at low dose rate), prescription location (apex versus base), plaque margin, and maximum allowed plaque size [9].

Real-world practice patterns generally adhere to guidelines yet vary among institutions. For instance, a survey among nine high-volume institutions utilizing eye plaque brachytherapy overall found good agreement in practice patterns. The survey reported clinical margins and planning systems to be similar among institutions, while prescription dose, dose rates, and dosimetry varied [11].

Dose De-escalation

85 Gy has been the standard dose prescription for iodine-125 plaque brachytherapy as it was the minimum dose prescribed in COMS report No. 19 (2002), the first prospective randomized clinical trial comparing iodine-125 plaque brachytherapy with enucleation for patients with medium-sized tumors. Dose modifications may be appropriate to account for different tumor sizes. Dose de-escalation may result in better visual and treatment convocation outcomes without compromising tumor control, especially with smaller tumors. Perez et al. reported their analysis of dose on disease control and visual outcomes with iodine-125 plaques and suggested a dose to tumor apex <85 Gy especially for tumors <5 mm in height [12]. Prospective dose de-escalation trials using ruthenium-106 brachytherapy, proton beam, and gamma knife radiosurgery have demonstrated favorable results. For iodine-125, dose de-escalation has yet to be investigated via randomized, prospective trials [13].

Radionuclides in Plaque Brachytherapy

Iodine-125 is the most commonly utilized isotope for eye plaque treatment, particularly in North America. This may be a consequence of its use in COMS trials establishing eye plaque brachytherapy as a treatment modality. Ruthenium-106 has been long established in Europe for eye plaque brachytherapy. As a beta-emitter, ruthenium-106 has limited dose penetration and generally demonstrates a lower ocular toxicity profile compared to iodine-125. As a consequence, it is well suited for smaller tumors [14] although limited to treatment of tumors <6 mm in thickness. Some centers performing ruthenium-106 plaque brachytherapy have also added transpupillary thermotherapy to supplement coverage for larger tumors [15]. Ruthenium-106 may also be utilized for treatment of juxtapapillary and circumpapillary choroidal melanoma due to sharper dose falloff. Palladium-103 has also been implicated to deliver lower radiation dose to normal ocular structures and more irradiation to tumor compared to iodine-125, which has translated to improved long-term visual acuity outcomes and reduced risk of radiation retinopathy [16, 17]. Other less commonly utilized radionuclides include strontium-90 and cesium-131. Table 9.1 summarizes key features of common radionuclides used in plaque brachytherapy (Fig. 9.1) [8].

Table 9.1 Common isotopes in plaque brachytherapy for uveal melanoma

Mode of emission	Half life	Regional use	Key facts
Photon			
Iodine-125	59.4 days	North America, Europe	Most commonly utilized isotope overall. Isotope used in COMS studies.
Palladium-103	17.0 days	North America	Lower dose to normal ocular structures compared to iodine-125, per some studies.
Cesium-131	9.7 days	–	–
Beta			
Ruthenium-106	371.8 days	Europe, Japan, Russia	Limited dose penetration and sharper dose fall off. Suitable for smaller tumors.
Strontium-90	28.8 years	Russia	–

Modified from The American Brachytherapy Society consensus guidelines for plaque brachytherapy of uveal melanoma, 2014

Particle/Proton Beam Radiotherapy

Indications for Use

Particle/proton beam radiotherapy is also commonly used in definitive treatment of uveal melanoma. The modality is additionally appropriate for tumors near the optic nerve/disc or macula (for which plaque brachytherapy may have difficulty balancing adequate coverage versus toxicity to surrounding tissue), for tumors with greater thickness and larger diameter (for which plaques cannot adequately penetrate depth or cover spatially), for tumors under the orbital muscles, after margin positive enucleation, and for intraocular/orbital recurrence.

Compared to plaque brachytherapy, there are a few drawbacks of particle/proton beam therapy. First, the eye can move independently of the external beam, which creates an uncertainty in dose deposition. Often, the patient is asked to fixate the eye at a specific point such that the intraocular target remains within the proton beam path. Secondly, the anterior entry dose tends to result in more anterior segment complications and is particularly a challenge for posteriorly located tumors. Finally, particle/proton beam therapy is more expensive.

Treatment Planning Considerations

Tantalum clips are utilized as fiducial markers for image guidance and verification. Volumetric planning with CT and/or MRI and selection of gaze angle, field collimation, and beam depth/width are performed to maximize delivery to the target volume and minimize dose to organs at risk. For proton beam therapy, general prescription is 50–70 CGyE in five fractions over 7–10 days. For carbon ions, dose prescription is 60–85 CGyE in five fractions [9].

Photon-Based Radiotherapy

Stereotactic Radiosurgery/Radiotherapy

Stereotactic radiosurgery (SRS) and stereotactic radiotherapy (SRT) are less often used radiation modalities in uveal melanoma, and few prospective studies have assessed efficacy and safety. Considerations for treatment planning include mobile intraocular target volume, as well as a greater dose inhomogeneity associated with photon-based planning when compared to protons [18]. The dose prescription is 18–45 Gy for single fraction and 45–70 Gy for multi-fraction treatment [9].

Conventional Photon Therapy

Photon beam radiotherapy has a role in the adjuvant setting following surgery for orbital involvement, particularly for patients at risk for local recurrence (i.e., positive margins, suspected residual subclinical disease) or regional recurrence. Adjuvant dosing is generally 20–30 Gy in five fractions. Additionally, photon beam radiotherapy is utilized for palliation for symptomatic distant metastases.

Table 9.2 summarizes key differences between the aforementioned radiotherapy treatment modalities (Table 9.2) [8].

Table 9.2 Comparison of plaque, proton, and stereotactic radiation therapy

Plaque brachytherapy	Proton/particle beam therapy	Stereotactic radiosurgery/ stereotactic radiotherapy
Mobile radiation field	Static radiation field	Static radiation field
Continuous low dose treatment	Daily high dose fractions	Daily high dose fractions
85 Gy for iodine-125 COMS plaques	50–70 CGyE in five fractions (protons)	18–45 Gy (single fraction treatment)
Generally 60–100 Gy at low dose rate over 3–7 days	60–85 CGyE in five fractions (protons)	45–70 Gy (multi-fraction treatment)
Most common and established	Additionally appropriate for tumors near optic disc/macula, larger tumors	Least used and studied
Size limitation	More expensive	Greater dose inhomogeneity when compared to protons
	More anterior segment complications	

Modified from The American Brachytherapy Society consensus guidelines for plaque brachytherapy of uveal melanoma, 2014

Outcomes Following Radiotherapy for Uveal Melanoma

Following definitive radiotherapy, the treated tumor tends to regress slowly over time. Reporting outcomes is challenging because, among different institutions, even with utilizing the same radiation modality, the method of plaque placement, radiotherapy planning, follow-up at reporting, and use of actuarial estimates can vary widely. We present below reported outcome metrics from a review of the literature as available at the time of this publication.

Local Control

Overall, for small to medium tumors, local control (LC) is in the range of 90–100% with 5-year overall survival (OS) >80%. For large tumors, LC is roughly 70% with 5-year OS about 70% [19]. Factors cited to impact local control include larger tumor dimension, extrascleral extension, close proximity to optic disc, lower radiation dose to tumor apex, lower radiation dose rate to tumor apex and base, and longer overall treatment time [1, 20].

Local Control Following Plaque Brachytherapy

Several reports in the literature report long-term outcomes for iodine-125 eye plaque brachytherapy for uveal melanoma. Local control rates upon literature review are reported in the range of 86.9–96% at 2–3 years, 80–98.3% at 5 years, and 93–94.4% at 8–10 years [21].

Image-guided brachytherapy at the time of plaque placement has been investigated and cited to be important in helping to decrease local failure rates. Tann et al. reported on early outcomes for uveal melanoma patients treated with iodine-125 brachytherapy utilizing intraoperative ultrasound guidance. The local control rate was 100% at a median follow-up of 21.6 months [22]. A 100% local control rate has since been maintained at a median follow-up of 36 months, for a larger patient cohort spanning across different stages, sizes, and GEP complications. Intraoperative image guidance likely contributed to accurate plaque placement, as evidenced by adequate coverage metrics (mean V85Gy of 96.8%) and image-guided repositioning utilized in almost one in six plaques [21].

A few studies have compared iodine-125 to other isotopes in eye plaque brachytherapy for uveal melanoma with respect to outcomes and toxicities. Takiar et al. performed a comparative analysis of iodine-125 and Ru-106, with actuarial 5-year local control at 83% for iodine-125 and at 97% for Ru-106 [14]. Leonard et al. reported on their institutional experience treating uveal melanoma with iodine-125, Ru-106, and Cs-131. In their literature review for which can be referenced, the

5-year local control rate for Ru-106 averaged over five select studies was 91.5%, and local control rate for Pd-103 averaged over two studies was 96.3% [23].

Local Control Following Particle/Proton Beam Therapy

Outcomes for proton and helium particle radiotherapy as modalities to treat uveal melanoma are also reported in the literature. Lin et al. examined the National Cancer Database in comparing proton beam radiotherapy versus episcleral plaque brachytherapy and found that 5-year local control rates for proton therapy ranged from 85% to 96%. They also concluded that proton beam radiotherapy reported inferior overall survival when compared to episcleral plaque brachytherapy (51% versus 77%, respectively) which was statistically significant [24]. Verma et al. sought to review outcomes of proton radiotherapy for uveal melanoma, drawing from 14 studies from 10 institutions. They found that 5-year local control rates exceeded 90%, which persisted at 10 and 15 years [25]. Mishra et al. reported on a prospective trial comparing helium particle radiotherapy to plaque treatment. Local control for particle versus plaque treatment was 100% versus 84% at 5 years and 98 versus 79% at 12 years ($P = 0.0006$). They concluded that helium particle therapy resulted in significantly improved local control, eye preservation, and disease-free survival [26].

Distant Metastasis

Overall for plaque brachytherapy and proton beam therapy, 5-year distant metastasis rate is 16–20% [19]. Larger tumor size is a risk factor for distant metastasis; for small choroidal melanoma treated with plaque brachytherapy, a study reported each millimeter of increasing thickness and diameter contributed risk for metastatic disease [27].

Local tumor recurrence is associated with a significantly higher risk of distant metastasis. Thus, effective initial treatment and long-term surveillance of treated uveal melanoma are necessary [1, 20]. Effective local treatment may eradicate locally aggressive disease prior to development of micrometastases, particularly in smaller tumors or tumors in transition from a less aggressive genotype to a more aggressive one.

The key prognostic factor for metastasis in uveal melanoma is the GEP (gene expression profile) class. This classification was developed to distinguish between uveal melanomas that have a low metastatic risk (class 1 tumors) versus high metastatic risk (class 2 tumors) [28]. GEP class has been validated as the strongest prognostic factor for metastasis, over TMN classification and the previous gold standard, chromosome 3 testing [29, 30].

Summary

Radiotherapy has an established role in multiple settings in the treatment of uveal melanoma, initially emerging as an effective treatment option that can spare both globe and visual function. Delivery modalities include eye plaque brachytherapy, proton/particle beam therapy, and photon-based treatments, each of which has select indications and limitations. Two principal goals in radiotherapy involve achieving excellent tumor coverage and control while sparing surrounding structures to preserve vision and quality of life.

References

1. Gallie BL, Simpson ER, Saakyan S, Amiryan A, Valskiy V, Finger PT, et al. Local recurrence significantly increases the risk of metastatic uveal melanoma. *Ophthalmology*. 2016;123(1):86–91. Available from: <http://www.sciencedirect.com/science/article/pii/S0161642015010362>.
2. Vrabec TR, Augsburger JJ, Gamel JW, Brady LW, Hernandez C, Woodleigh R. Impact of local tumor relapse on patient survival after Cobalt 60 plaque radiotherapy. *Ophthalmology*. 1991;98(6):984–8. Available from: [https://doi.org/10.1016/S0161-6420\(91\)32193-6](https://doi.org/10.1016/S0161-6420(91)32193-6).
3. Edge SB, Edge SB. Cancer. In: *AJC on AJCC cancer staging manual*. 8th ed. New York: Springer; 2017.
4. Mortality in patients with small choroidal melanoma. COMS report no. 4. The Collaborative Ocular Melanoma Study Group. *Arch Ophthalmol*. 1997;115(7):886–93.
5. Factors predictive of growth and treatment of small choroidal melanoma: COMS report no. 5. The Collaborative Ocular Melanoma Study Group. *Arch Ophthalmol*. 1997;115(12):1537–44.
6. Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: V. twelve-year mortality rates and prognostic factors: COMS Report No. 28. *Arch Ophthalmol*. 2006;124(12):1684–93. Available from: <https://doi.org/10.1001/archophth.124.12.1684>.
7. Hawkins BS. The collaborative ocular melanoma study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma: IV. Ten-year mortality findings and prognostic factors. COMS report number 24. *Am J Ophthalmol*. 2004;138(6):936–51.
8. American Brachytherapy Society - Ophthalmic Oncology Task Force. The American Brachytherapy Society consensus guidelines for plaque brachytherapy of uveal melanoma and retinoblastoma. *Brachytherapy*. 2014;13(1):1–14.
9. National Comprehensive Cancer Network. Uveal Melanoma (Version 2.2020). [Internet]. Available from: https://www.nccn.org/professionals/physician_gls/pdf/uveal.pdf.
10. Chiu-Tsao S-T, Astrahan MA, Finger PT, Followill DS, Meigooni AS, Melhus CS, et al. Dosimetry of (125)I and (103)Pd COMS eye plaques for intraocular tumors: report of Task Group 129 by the AAPM and ABS. *Med Phys*. 2012;39(10):6161–84.
11. Binder C, Mruthunjaya P, Scheffler AC, Seider MI, Crilly R, Hung A, et al. Practice patterns for the treatment of uveal melanoma with Iodine-125 plaque brachytherapy: ocular oncology study consortium report 5. *Ocul Oncol Pathol*. 2020;6(3):210–8. Available from: <https://www.karger.com/DOI/10.1159/000504312>.
12. Perez BA, Mettu P, Vajzovic L, Rivera D, Alkaissi A, Steffey BA, et al. Uveal melanoma treated with iodine-125 episcleral plaque: an analysis of dose on disease control and visual outcomes. *Int J Radiat Oncol Biol Phys*. 2014;89(1):127–36.

13. Le Bhan A, Patel R, Jennelle R, Berry JL. Evidence for dose de-escalation in brachytherapy for choroidal melanoma. *Adv Ophthalmol Optom*. 2018;3(1):139–53. Available from: <https://doi.org/10.1016/j.yao.2018.04.008>.
14. Takiar V, Voong KR, Gombos DS, Mourtada F, Rechner LA, Lawyer AA, et al. A choice of radionuclide: comparative outcomes and toxicity of ruthenium-106 and iodine-125 in the definitive treatment of uveal melanoma. *Pract Radiat Oncol*. 2015;5(3):e169–76. Available from: <http://www.sciencedirect.com/science/article/pii/S1879850014002483>.
15. Kreusel K-M, Bechrakis N, Riese J, Krause L, Wachtlin J, Foerster MH. Combined brachytherapy and transpupillary thermotherapy for large choroidal melanoma: tumor regression and early complications. *Graefes Arch Clin Exp Ophthalmol*. 2006;244(12):1575–80. Available from: <https://doi.org/10.1007/s00417-006-0357-8>.
16. Finger PT, Zhou D, Kalach N, Semenova E, Choi W. 103Pd versus 125I ophthalmic plaque brachytherapy: preoperative comparative radiation dosimetry for 319 uveal melanomas. *J Radiat Oncol*. 2014;3(4):409–16. Available from: <https://doi.org/10.1007/s13566-014-0149-4>.
17. Patel KR, Prabhu RS, Switchenko JM, Chowdhary M, Craven C, Mendoza P, et al. Visual acuity, oncologic, and toxicity outcomes with 103Pd vs. 125I plaque treatment for choroidal melanoma. *Brachytherapy*. 2017;16(3):646–53. Available from: <http://www.sciencedirect.com/science/article/pii/S1538472117300181>.
18. Weber DC, Bogner J, Verwey J, Georg D, Dieckmann K, Escudé L, et al. Proton beam radiotherapy versus fractionated stereotactic radiotherapy for uveal melanomas: a comparative study. *Int J Radiat Oncol Biol Phys*. 2005;63(2):373–84.
19. Juloori A, Yu JS, Chao ST. Glioblastoma. In: Ward Rahul D, Tendulkar RD, Videtic GM, editors. *Essentials of clinical radiation oncology*. 1st ed. New York: Springer Publishing Company. Available from: <https://connect.springerpub.com/content/book/978-0-8261-6855-9>.
20. Quivey JM, Augsburger J, Snelling L, Brady LW. 125I plaque therapy for uveal melanoma: Analysis of the impact of time and dose factors on local control. *Cancer*. 1996;77(11):2356–62. Available from: [https://doi.org/10.1002/\(SICI\)1097-0142\(19960601\)77:11%3C2356::AID-CNCR26%3E3.0.CO](https://doi.org/10.1002/(SICI)1097-0142(19960601)77:11%3C2356::AID-CNCR26%3E3.0.CO).
21. Wong AJ, Teh BS, Nguyen BT, Pino R, Bretana ME, Bernicker EH, Chevez-Barrios P, Butler EB, Scheffler AC. Three-Year Outcomes of Uveal Melanoma Treated with Intraoperative Ultrasound Guided I-125 Brachytherapy Using Custom Built Eye Plaques. Manuscript submitted for publication.
22. Tann AW, Teh BS, Scarboro SB, Lewis GD, Bretana ME, Croft PC, et al. Early outcomes of uveal melanoma treated with intraoperative ultrasound guided brachytherapy using custom built plaques. *Pract Radiat Oncol*. 2017;7(4):e275–82. Available from: <https://doi.org/10.1016/j.prro.2017.01.002>.
23. Leonard KL, Gagne NL, Mignano JE, Duker JS, Bannon EA, Rivard MJ. A 17-year retrospective study of institutional results for eye plaque brachytherapy of uveal melanoma using 125I, 103Pd, and 131Cs and historical perspective. *Brachytherapy*. 2011;10(4):331–9. Available from: <http://www.sciencedirect.com/science/article/pii/S1538472111000092>.
24. Lin AJ, Rao YJ, Acharya S, Schwarz J, Rao PK, Grigsby P. Patterns of care and outcomes of proton and eye plaque brachytherapy for uveal melanoma: review of the National Cancer Database. *Brachytherapy*. 2017;16(6):1225–31. Available from: <http://www.sciencedirect.com/science/article/pii/S1538472117304191>.
25. Verma V, Mehta MP. Clinical outcomes of proton radiotherapy for uveal melanoma. *Clin Oncol*. 2016;28(8):e17–27. Available from: <http://www.sciencedirect.com/science/article/pii/S0936655516000637>.
26. Mishra KK, Quivey JM, Daftari IK, Weinberg V, Cole TB, Patel K, et al. Long-term results of the UCSF-LBNL randomized trial: charged particle with helium ion versus Iodine-125 plaque therapy for choroidal and ciliary body melanoma. *Int J Radiat Oncol*. 2015;92(2):376–83. Available from: <http://www.sciencedirect.com/science/article/pii/S0360301615001212>.
27. Shields CL, Sioufi K, Srinivasan A, Di Nicola M, Masoomian B, Barna LE, et al. Visual outcome and millimeter incremental risk of metastasis in 1780 patients with small choro-

- dal melanoma managed by plaque radiotherapy. *JAMA Ophthalmol.* 2018;136(12):1325–33. Available from: <https://doi.org/10.1001/jamaophthalmol.2018.3881>.
28. Harbour JW. A prognostic test to predict the risk of metastasis in uveal melanoma based on a 15-gene expression profile. *Methods Mol Biol.* 2014;1102:427–40.
 29. Corrêa ZM, Augsburger JJ. Independent prognostic significance of gene expression profile class and largest basal diameter of posterior uveal melanomas. *Am J Ophthalmol.* 2016;162:20–27. e1. Available from: <http://www.sciencedirect.com/science/article/pii/S0002939415007072>.
 30. Onken MD, Worley LA, Char DH, Augsburger JJ, Correa ZM, Nudleman E, et al. Collaborative ocular oncology group report number 1: prospective validation of a multi-gene prognostic assay in uveal melanoma. *Ophthalmology.* 2012;119(8):1596–603. Available from: <http://www.sciencedirect.com/science/article/pii/S0161642012001376>.